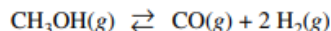


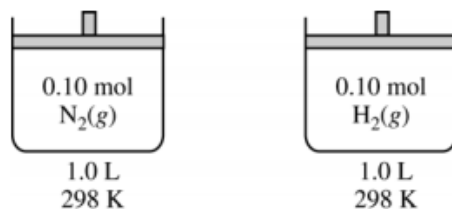
2011 AP[®] CHEMISTRY FREE-RESPONSE QUESTIONS (Form B)

2. An 8.55 mol sample of methanol, CH_3OH , is placed in a 15.0 L evacuated rigid tank and heated to 327°C . At that temperature, all of the methanol is vaporized and some of the methanol decomposes to form carbon monoxide gas and hydrogen gas, as represented in the equation below.



- (a) The reaction mixture contains 6.30 mol of $\text{CO}(g)$ at equilibrium at 327°C .
- Calculate the number of moles of $\text{H}_2(g)$ in the tank.
 - Calculate the number of grams of $\text{CH}_3\text{OH}(g)$ remaining in the tank.
 - Calculate the mole fraction of $\text{H}_2(g)$ in the tank.
 - Calculate the total pressure, in atm, in the tank at 327°C .
- (b) Consider the three gases in the tank at 327°C : $\text{CH}_3\text{OH}(g)$, $\text{CO}(g)$, and $\text{H}_2(g)$.
- How do the average kinetic energies of the molecules of the gases compare? Explain.
 - Which gas has the highest average molecular speed? Explain.
- (c) The tank is cooled to 25°C , which is well below the boiling point of methanol. It is found that small amounts of $\text{H}_2(g)$ and $\text{CO}(g)$ have dissolved in the liquid CH_3OH . Which of the two gases would you expect to be more soluble in methanol at 25°C ? Justify your answer.

2005 AP[®] CHEMISTRY FREE-RESPONSE QUESTIONS (Form B)



6. Consider two containers of volume 1.0 L at 298 K, as shown above. One container holds 0.10 mol $\text{N}_2(g)$ and the other holds 0.10 mol $\text{H}_2(g)$. The average kinetic energy of the $\text{N}_2(g)$ molecules is $6.2 \times 10^{-21} \text{ J}$. Assume that the $\text{N}_2(g)$ and the $\text{H}_2(g)$ exhibit ideal behavior.
- Is the pressure in the container holding the $\text{H}_2(g)$ less than, greater than, or equal to the pressure in the container holding the $\text{N}_2(g)$? Justify your answer.
 - What is the average kinetic energy of the $\text{H}_2(g)$ molecules?
 - The molecules of which gas, N_2 or H_2 , have the greater average speed? Justify your answer.
 - What change could be made that would decrease the average kinetic energy of the $\text{N}_2(g)$ molecules in the container?
 - If the volume of the container holding the $\text{H}_2(g)$ was decreased to 0.50 L at 298 K, what would be the change in each of the following variables? In each case, justify your answer.
 - The pressure within the container
 - The average speed of the $\text{H}_2(g)$ molecules

From 2009 #2

2. A student was assigned the task of determining the molar mass of an unknown gas. The student measured the mass of a sealed 843 mL rigid flask that contained dry air. The student then flushed the flask with the unknown gas, resealed it, and measured the mass again. Both the air and the unknown gas were at 23.0°C and 750. torr. The data for the experiment are shown in the table below.

Volume of sealed flask	843 mL
Mass of sealed flask and dry air	157.70 g
Mass of sealed flask and unknown gas	158.08 g

- (a) Calculate the mass, in grams, of the dry air that was in the sealed flask. (The density of dry air is 1.18 g L^{-1} at 23.0°C and 750. torr.)
- (b) Calculate the mass, in grams, of the sealed flask itself (i.e., if it had no air in it).
- (c) Calculate the mass, in grams, of the unknown gas that was added to the sealed flask.
- (d) Using the information above, calculate the value of the molar mass of the unknown gas.

After the experiment was completed, the instructor informed the student that the unknown gas was carbon dioxide (44.0 g mol^{-1}).

- (e) Calculate the percent error in the value of the molar mass calculated in part (d).
- (f) For each of the following two possible occurrences, indicate whether it by itself could have been responsible for the error in the student's experimental result. You need not include any calculations with your answer. For each of the possible occurrences, justify your answer.

Occurrence 1: The flask was incompletely flushed with $\text{CO}_2(\text{g})$, resulting in some dry air remaining in the flask.

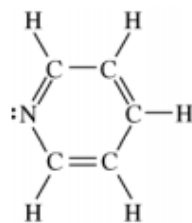
Occurrence 2: The temperature of the air was 23.0°C, but the temperature of the $\text{CO}_2(\text{g})$ was lower than the reported 23.0°C.

- (g) Describe the steps of a laboratory method that the student could use to verify that the volume of the rigid flask is 843 mL at 23.0°C. You need not include any calculations with your answer.

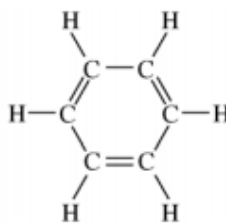
2008 AP[®] CHEMISTRY FREE-RESPONSE QUESTIONS

6. Answer the following questions by using principles of molecular structure and intermolecular forces.

- (a) Structures of the pyridine molecule and the benzene molecule are shown below. Pyridine is soluble in water, whereas benzene is not soluble in water. Account for the difference in solubility. You must discuss both of the substances in your answer.

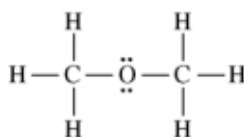


Pyridine

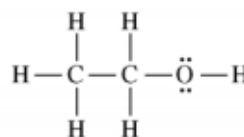


Benzene

- (b) Structures of the dimethyl ether molecule and the ethanol molecule are shown below. The normal boiling point of dimethyl ether is 250 K, whereas the normal boiling point of ethanol is 351 K. Account for the difference in boiling points. You must discuss both of the substances in your answer.



Dimethyl Ether



Ethanol

- (c) SO_2 melts at 201 K, whereas SiO_2 melts at 1,883 K. Account for the difference in melting points. You must discuss both of the substances in your answer.
- (d) The normal boiling point of $\text{Cl}_2(l)$ (238 K) is higher than the normal boiling point of $\text{HCl}(l)$ (188 K). Account for the difference in normal boiling points based on the types of intermolecular forces in the substances. You must discuss both of the substances in your answer.